

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 HSC ASSESSMENT TASK 2

MARCH 2008

MATHEMATICS

Extension 1

Time Allowed: 70 minutes

Instructions:

- Attempt all questions
- Start each question on a new page
- Show all necessary working
- The marks for each question are indicated next to the question
- Marks may be deducted for careless or badly arranged work
- Marks indicated are a guide only and may be varied if necessary

Name: _____ Teacher: _____

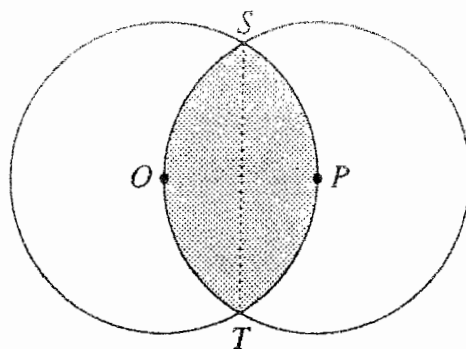
Question 1	Question 2	Question 3	Question 4	Question 5	Total

QUESTION 1 (10 Marks)

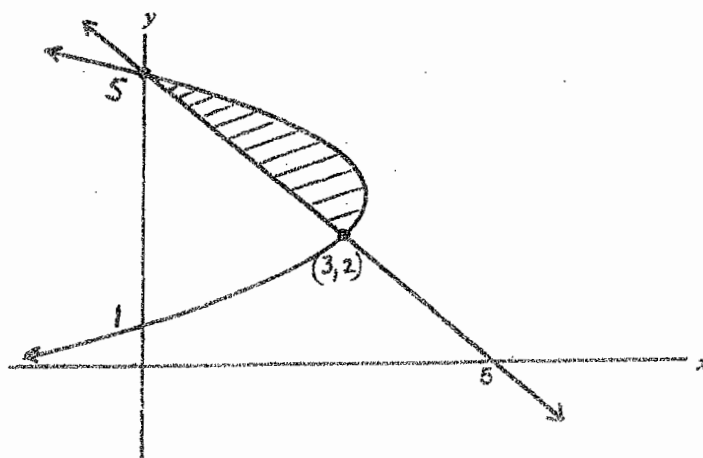
- a) Find the gradient of the tangent to the curve $y = \cos^3 x$ at $x = \frac{\pi}{6}$ **2**
- b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{4x}$ **1**
- c) Write a primitive for $(5 - 2x)^4$ **2**
- d) Find $\int \frac{x \, dx}{(1+x^2)^2}$ by first differentiating $\frac{x^2}{1+x^2}$ **2**
- e) Evaluate $\int_0^{\sqrt{3}} \frac{x \, dx}{\sqrt{1+x^2}}$ using the substitution $u = 1+x^2$ **3**

QUESTION 2 (10 Marks)

- a) The points O and P in the plane are d cm apart. A circle centre O is drawn to pass through P , and another circle centre P is drawn to pass through O . The two circles meet at S and T , as in the diagram.



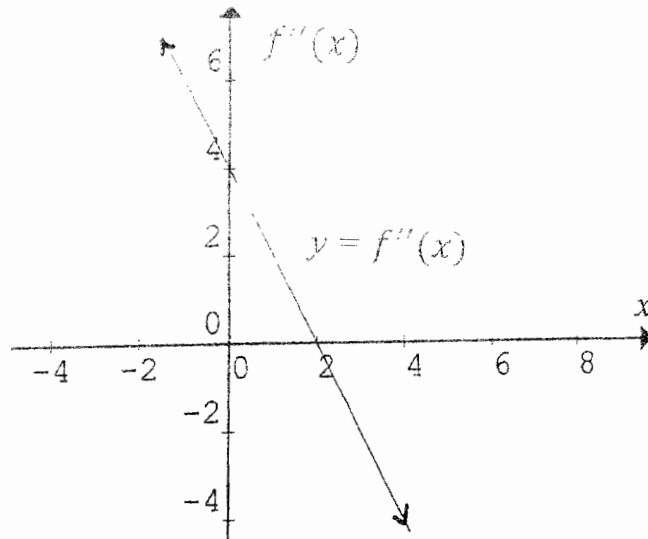
- i) Explain why angle SOT is $\frac{2\pi}{3}$ 2
 - ii) Hence find the exact area of the shaded region in terms of d 2
- b) The diagram shows the curve $x = 6y - 5 - y^2$ and the line $x + y = 5$ 3
- The two graphs intersect at $(0, 5)$ and $(3, 2)$



Determine the magnitude of the shaded area

- c) For the curve $y = x^5 - 80x$, $\frac{d^2y}{dx^2} = 0$ at $(0,0)$. Is $(0,0)$ a point of inflexion? Justify your answer. 3

QUESTION 3 (10 Marks)



This is the graph of $y = f''(x)$

- i) Find the equation of $f'(x)$ if there is a stationary point at $(1,4)$ 2
- ii) What is the nature of the stationary point at $(1,4)$? Give a reason. 1
- b)
- i) Sketch on the same diagram the graphs of $y = 2 \sin x$ and $y = \cos 2x$ for $0 \leq x \leq 2\pi$ 3
- ii) Use your graph or otherwise determine a value for d , where d is an integer, so that the equation $2 \sin x - \cos 2x = d$ has 4 solutions in the interval $0 \leq x \leq 2\pi$ 1
- c) Find $\int x^3 (x^2 + 1)^2 dx$ by using the substitution $u = x^2 + 1$ 3

QUESTION 4 (10 Marks)

a) For the curve $y = \frac{x^2}{1+x}$

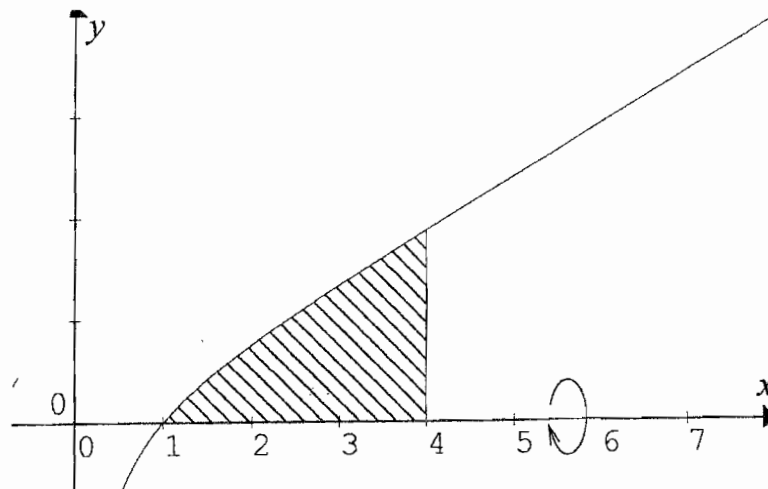
i) Find the co-ordinates of the stationary points and determine their nature. 3

ii) Given $y = \frac{x^2}{1+x}$ can be written as $y = x - 1 + \frac{1}{x+1}$

Write down the equations of any asymptotes 2

iii) Sketch the curve showing the stationary points and the asymptotes 2

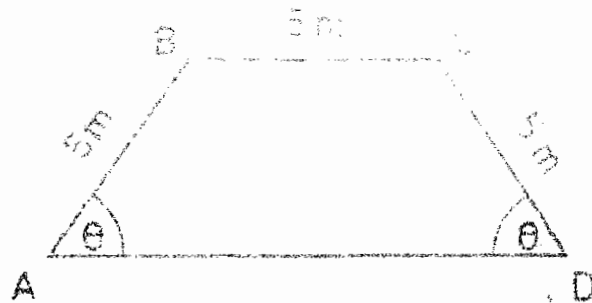
b) 3



The shaded region in the diagram is bounded by the curve $y = x - \frac{1}{x}$, the x -axis and the line $x = 4$. Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis.

QUESTION 5 (10 Marks)

a)



In a quadrilateral $ABCD$, BC is parallel to AD , the sides AB , BC , CD are each 5m long and the angles BAD , ADC each have size θ , as shown in the diagram:

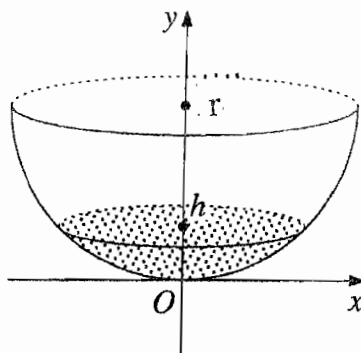
- i) Show that the area of the trapezium is given by the formula 2

$$\text{Area} = 25 \sin \theta (1 + \cos \theta)$$

- ii) Hence find the value of θ for which this area is a maximum 4

- b) A hemi-spherical bowl is formed by rotating the semi-circle 4

$y = r - \sqrt{r^2 - x^2}$ about the y -axis. The bowl contains water up to the height h where $0 < h < r$.



Show that the volume of water in the bowl is $\frac{\pi h^2(3r-h)}{3}$

① Question 1

$$a) y = \cos^3 x$$

$$y' = -3 \sin x \cdot \cos^2 x$$

$$x = \pi/6$$

$$y' = -3 \cdot \frac{1}{2} \cdot \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= -9/8$$

$$b) \lim_{x \rightarrow 0} \frac{\sin 5x}{4x} = 5/4$$

$$c) \frac{-(5-2x)^5}{10} + C$$

$$d) \frac{d}{dx} \left(\frac{x^2}{1+x^2} \right) = \frac{(1+x^2) \cdot 2x - x^2 \cdot 2x}{(1+x^2)^2}$$

$$= \frac{2x}{(1+x^2)^2}$$

$$\therefore \int \frac{x}{(1+x^2)^2} dx = \frac{1}{2} \frac{x^2}{1+x^2} + C$$

$$e) u = 1+x^2 \rightarrow du = 2x dx$$

$$x = \sqrt{3}$$

$$u = 4$$

$$x = 0$$

$$u = 1$$

$$\therefore = \frac{1}{2} \int_0^4 \frac{du}{u^{1/2}}$$

$$= \frac{1}{2} \int_0^4 u^{-1/2} du$$

$$= \frac{1}{2} \left[2u^{1/2} \right]_1^4$$

$$= 1$$

Question 1

$$y = \frac{1}{2}x^2 - 5x + 12$$

$$x = 2 \text{ to } x = 5$$

$$y = 2 \text{ to } y = 12$$

$$1) \text{ Area} = \int_2^5 \left[\frac{1}{2}x^2 - 5x + 12 - 2 \right] dx$$

$$= \int_2^5 \left[\frac{1}{2}x^2 - 5x + 10 \right] dx$$

$$b) \text{ Area} = \int_2^5 (6y - 5 - y^2) dy - \frac{1}{2} \times 3 \times 3$$

$$= \left[3y^2 - 5y - \frac{y^3}{3} \right]_2^5 - \frac{9}{2}$$

$$= \left[75 - 25 - \frac{125}{3} \right] - \left[12 - 10 - \frac{8}{3} \right] - \frac{9}{2}$$

$$= 8\frac{1}{3} - (-\frac{2}{3}) - \frac{9}{2}$$

$$= 4\frac{1}{2}$$

$$c) y' = 5x^4 - 80$$

$$y' = 0 \text{ for stationary pt.}$$

$$\therefore 5x^4 - 80 = 0$$

$$5[x^4 - 16] = 0$$

$$\therefore x = \pm 2$$

$$\therefore \text{stationary pts at } x = \pm 2$$

$\therefore (0,0)$ is a point of inflection as it is between two stationary points on a continuous curve.

or

$$y'' = 20x^3$$

x	-1	0	1
y''	-	0	+

curve changes concavity on either side of $(0,0)$

$$\begin{aligned}
 &= r^2 - [r^2 - 2xy + y^2] \\
 &= 2xy - y^2
 \end{aligned}$$

$$\text{Volume} = \pi \int_0^h 2xy - y^2 dy$$

$$= \pi \left[xy^2 - \frac{y^3}{3} \right]_0^h$$

$$= \pi \left[rh^2 - \frac{h^3}{3} \right]$$

$$= \pi \left[\frac{3rh^2 - h^3}{3} \right]$$

$$= \frac{\pi h^2 [3r - h]}{3}$$

Teacher's Sign

Student's Name

Question 3

$$a) \quad f(x) = 2x^2 + 4x - 3$$

$$f'(x) = 4x + 4$$

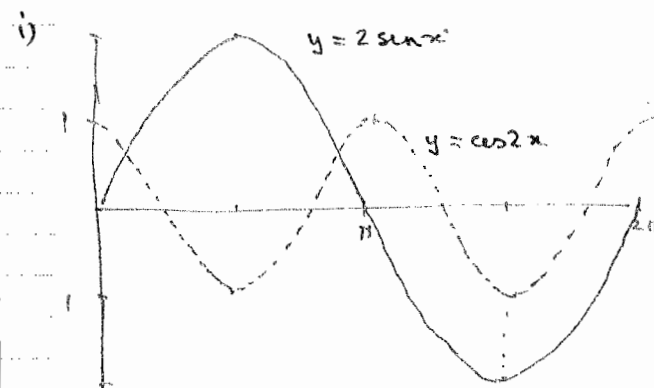
$$\text{at } (1, 4) \quad 0 = 4 + 4$$

$$c = -3$$

$$\therefore f'(x) = -x^2 + 4x - 3$$

$$ii) \quad \text{at } x = 1 \quad f''(x) > 0$$

$\therefore$ minimum turning point



$$ii) \quad 2 \sin x = \cos 2x + d$$

$$\therefore d = -1$$

$$c) \quad u = x^2 + 1 \quad \Rightarrow du = 2x dx$$

$$x^2 = u - 1$$

$$\therefore \int x^3 (x^2 + 1)^2 dx = \int x^2 \cdot x \cdot (x^2 + 1)^2 dx$$

$$= \int (u - 1) \cdot u^2 \cdot \frac{du}{2}$$

$$= \frac{1}{2} \int u^3 - u du$$

$$= \frac{1}{2} \left[\frac{u^4}{4} - \frac{u^3}{3} \right] + c$$

$$= \frac{(x^2 + 1)^4}{8} - \frac{(x^2 + 1)^3}{6} + c$$

Question 4

$$a) \quad i) \quad y' = \frac{(1+x) \cdot 2x}{(1+x)^2}$$

$$= \frac{2x+x^2}{(1+x)^2}$$

 $y' = 0$ for stationary point

$$\therefore 2x+x^2=0$$

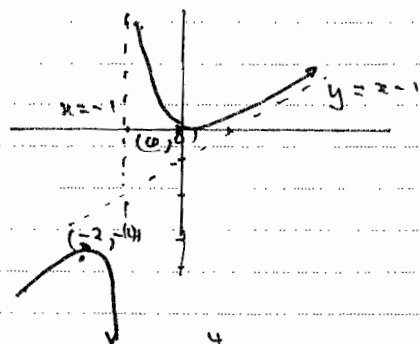
$$x(2+x)=0$$

$$x = -2 \text{ or } 0$$

x	-3	-2	-1.5	-1	0	1
y'	+	0	-	-	0	+

 $\therefore$ maximum at $(-2, -4)$ minimum at $(0, 0)$

$$ii) \quad y = x-1 \text{ at } x = -1$$



$$\begin{aligned}
 b) \quad \text{Volume} &= \pi \int_{-1}^2 \left(x - \frac{1}{x}\right)^2 dx \\
 &= \pi \int_{-1}^2 \left(x^2 - 2 + \frac{1}{x^2}\right) dx \\
 &= \pi \left[\frac{x^3}{3} - 2x - \frac{1}{x} \right]_{-1}^2 \\
 &= \pi \left[\left(\frac{8}{3} - 4 - \frac{1}{2}\right) - \left(-\frac{1}{3} - 2 - 1\right) \right] \\
 &= \frac{63\pi}{4}
 \end{aligned}$$

Question 5

a)

ii)



$$h = 5 \sin \theta \text{ at } d = 5 \cos \theta$$

$$\begin{aligned}
 \therefore \text{Area} &= \frac{5 \sin \theta}{2} [5 + 5 + 10 \cos \theta] \\
 &= 25 \sin \theta [1 + \cos \theta]
 \end{aligned}$$

$$\begin{aligned}
 ii) \quad A' &= 25 [(1 + \cos \theta) \cos \theta + \sin \theta \cdot \sin \theta] \\
 &= 25 [\cos \theta + \cos^2 \theta + \sin^2 \theta] \\
 &= 25 [\cos \theta + \cos^2 \theta - [1 - \cos^2 \theta]] \\
 &= 25 [2 \cos^2 \theta + \cos \theta - 1]
 \end{aligned}$$

For maximum $A' = 0$

$$\begin{aligned}
 \therefore 25 [2 \cos^2 \theta + \cos \theta - 1] &= 0 \\
 25 [(2 \cos \theta - 1)(\cos \theta + 1)] &= 0
 \end{aligned}$$

$$\begin{aligned}
 \therefore \cos \theta &= \frac{1}{2} \text{ or } \cos \theta = -1 \text{ (out of domain)} \\
 \theta &= \frac{\pi}{3}
 \end{aligned}$$

θ	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
A'	+	0	0

 $\therefore$ max area when $\theta = \frac{\pi}{3}$